Soot-Driven Glacier Loss Simulation

A Monte Carlo cellular automata simulation of black carbon deposition, albedo feedback, and glacier survival under two emission scenarios

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1 Introduction

Mountain glaciers are losing mass at accelerating rates. One of the less obvious drivers of glacier retreat is black carbon (BC), or soot, emitted by factories, vehicles, and biomass burning. When BC particles land on snow and ice, they darken the surface and lower its albedo (the fraction of incoming sunlight that gets reflected). Darker ice absorbs more solar energy, melts faster, and as it melts, the soot concentrates on the shrinking surface rather than evaporating away. The result is a positive feedback loop where more melt leads to darker ice leads to even more melt.

Research on the Tibetan Plateau has shown that BC deposition accounts for roughly 13% of annual glacier-wide melting, with atmospheric BC alone reducing surface albedo by about 0.02 and driving roughly 9% of total melt [1]. Studies on North American glaciers found that albedo declined by 0.018–0.019 per decade between 2000 and 2019, and that the resulting increase in absorbed radiation explained 31–41% of increased melt rates across western North America and the Canadian Arctic [2]. Field experiments confirmed that soot reduces snow albedo in a dose-dependent way consistent with the SNICAR radiative transfer model, and that older, coarser-grained snow is more sensitive to soot contamination than fresh fine-grained snow [3].

The research question for this project is: **if local factories installed black carbon scrubbers that reduced soot emissions by 80%, how much longer would a mountain glacier survive compared to a business-as-usual scenario, and how would peak meltwater discharge change?**

To answer this, I built a Monte Carlo cellular automata (CA) simulation of a synthetic glacier on a 100×100 grid. The simulation tracks ice thickness, soot concentration, and albedo for each cell over monthly time steps. Two strategies are compared: Business as Usual (BAU) with current soot emission rates, and a Scrubbers scenario where factory emissions are reduced by 80%. By running 50 paired simulations per strategy over 40 years, I measure the distribution of glacier lifespans, volume trajectories, and peak meltwater discharge under each scenario.

2 Model Description

2.1 Grid Geometry

The glacier sits on a 100×100 grid where each cell represents a 30 m × 30 m area, spanning a total domain of 3 km × 3 km. The initial ice surface is generated as a 2-D Gaussian:

$$h(x, y) = H_{\text{peak}} \cdot \exp\left(-\frac{(x - x_0)^2}{2\sigma_x^2} - \frac{(y - y_0)^2}{2\sigma_y^2}\right) + \epsilon \tag{1}$$

where $H_{\text{peak}} = 120\,\text{m}$ is the peak ice thickness at the dome center, $\sigma_x = 30$ cells and $\sigma_y = 25$ cells define the spread (making the glacier slightly elongated, like a real valley glacier), and $\epsilon \sim \text{Uniform}(-2, +2)\,\text{m}$ adds surface roughness. Cells where the resulting thickness falls below 2 m are set to zero (bare rock). The initial glacier volume is $0.439\,\text{km}^3$ covering the full $9.00\,\text{km}^2$ domain.

2.2 Temperature and Solar Forcing

Monthly temperature at the glacier base elevation follows a sinusoidal seasonal cycle with a linear warming trend and stochastic noise:

$$T(t) = T_{\text{base}} + A_{\text{season}} \cdot \sin\left(\frac{2\pi(m-4)}{12}\right) + r_{\text{warm}} \cdot y + \mathcal{N}(0, \sigma_T) \tag{2}$$

where $T_{\text{base}} = -2.0\,^{\circ}\text{C}$ is the annual mean, $A_{\text{season}} = 10.0\,^{\circ}\text{C}$ is the seasonal amplitude (so summer peaks near $+8\,^{\circ}\text{C}$ and winter troughs near $-12\,^{\circ}\text{C}$), $r_{\text{warm}} = 0.03\,^{\circ}\text{C/yr}$ is the warming trend, m is the calendar month, y is the year, and $\sigma_T = 1.5\,^{\circ}\text{C}$ is the monthly noise. Peak temperature occurs in July (month 7).

Each cell's effective temperature is adjusted by a lapse rate of $-6.5\,^{\circ}\text{C/km}$ based on the cell's elevation. Since the glacier is synthetic, each cell's initial ice thickness serves as a proxy for its elevation on the mountain. Thicker cells are assumed to be higher up and therefore colder. The per-cell temperature becomes:

$$T_{\text{cell}}(t) = T(t) + \frac{\text{lapse\_rate} \cdot h_{\text{base}}}{1000} \tag{3}$$

Incoming shortwave radiation also varies seasonally:

$$S(t) = S_{\text{mean}} + S_{\text{amp}} \cdot \sin\left(\frac{2\pi(m-3)}{12}\right) \tag{4}$$

with $S_{\text{mean}} = 200\,\text{W/m}^2$ and $S_{\text{amp}} = 150\,\text{W/m}^2$, so summer receives roughly $350\,\text{W/m}^2$ and winter roughly $50\,\text{W/m}^2$.

2.3 Albedo Model

Each cell's albedo depends on whether it has fresh snow cover and how much soot has accumulated on its surface. The base albedo starts at 0.85 for freshly snow-covered ice and decays linearly at 0.08 per month toward a bare-ice floor of 0.55 as the snow ages:

$$\alpha_{\text{base}} = \max(\alpha_{\text{ice}},\ \alpha_{\text{snow}} - d_{\text{decay}} \cdot t_{\text{snow}}) \tag{5}$$

where $\alpha_{\text{snow}} = 0.85$, $\alpha_{\text{ice}} = 0.55$, $d_{\text{decay}} = 0.08/\text{month}$, and t_{snow} is the number of months since the last snowfall event.

Soot darkening is modeled with a log-linear relationship inspired by the SNICAR radiative transfer model [4, 3]. The albedo reduction from a BC concentration C (in ppb) is:

$$\Delta\alpha = K_{\text{BC}} \cdot \ln\left(1 + \frac{C}{C_{\text{ref}}}\right) \tag{6}$$

with $K_{\text{BC}} = 0.10$ and $C_{\text{ref}} = 10.0\,\text{ppb}$. The reference concentration is chosen so that 10 ppb of BC produces an albedo reduction of about 0.07, consistent with field measurements showing that 10–20 ppb of BC reduces snow albedo by several percent [3, 5]. The effective albedo is then:

$$\alpha_{\text{eff}} = \max(\alpha_{\min},\ \alpha_{\text{base}} - \Delta\alpha) \tag{7}$$

with $\alpha_{\min} = 0.15$ (rock albedo). Exposed rock cells are unconditionally set to 0.15.

2.4 Energy Balance Melt

Melt rate is computed from a simplified surface energy balance. Research on Himalayan and High Mountain Asia glaciers has shown that net shortwave radiation is the dominant energy input at high elevation [8]. The monthly ice loss for each cell is:

$$M = \max\left(0,\ \frac{S \cdot (1 - \alpha_{\text{eff}}) \cdot f_{\text{melt}}}{\rho_{\text{ice}} \cdot L_f}\right) \times 86400 \times 30 \tag{8}$$

where S is the incoming shortwave (W/m^2), $\rho_{\text{ice}} = 900\,\text{kg/m}^3$, $L_f = 334{,}000\,\text{J/kg}$ is the latent heat of fusion, and f_{melt} is a temperature-dependent factor ensuring no melting occurs below freezing:

$$f_{\text{melt}} = \max\left(0,\ \frac{T_{\text{cell}}}{T_{\text{scale}}}\right) \tag{9}$$

with $T_{\text{scale}} = 10\,^\circ\text{C}$. Multiplication by 86400×30 converts from meters per second to meters per month.

2.5 Soot Deposition and Concentration Feedback

BC is deposited onto all glaciated cells each month. Under BAU, the monthly deposition is drawn from $\mathcal{N}(5.0, 2.0)$ ppb (clipped at zero). Under the Scrubbers scenario, deposition is drawn from $\mathcal{N}(1.0, 0.5)$ ppb, representing an 80% emission reduction. These values are within the range of observed BC concentrations in Himalayan glacier snow, where measurements span 5–500 ppb depending on region and season [4, 7].

When ice melts, the soot deposited on the surface does not evaporate with the meltwater. Instead, it concentrates on the remaining ice. If a cell has thickness h before melt and loses Δh, the new BC concentration is:

$$C_{\text{new}} = C_{\text{old}} \cdot \frac{h}{\max(0.01,\ h - \Delta h)} + D \tag{10}$$

where D is the new atmospheric deposition. As the glacier thins, the same mass of soot covers less and less ice, driving albedo lower and lower. When a cell fully melts out (thickness reaches zero), its soot is flushed away with the meltwater.

2.6 Wind Redistribution

Each month, 15% of each cell's surface BC is moved one cell in the prevailing wind direction (default: east). This creates a spatial gradient where downwind cells accumulate more soot than upwind cells, adding spatial heterogeneity to the simulation.

2.7 Glacial Flow

Ice flows from higher cells to lower neighbors due to gravity. For each pair of adjacent cells, if the thickness difference exceeds 5 m, a fraction of the excess moves:

$$\Delta h_{\text{transfer}} = r_{\text{flow}} \cdot (h_A - h_B) \quad \text{if } h_A - h_B > h_{\text{threshold}} \tag{11}$$

with $r_{\text{flow}} = 0.02$ and $h_{\text{threshold}} = 5\,\text{m}$. This is applied to all four cardinal neighbors each month. It is a simplified version of the shallow ice approximation and causes the glacier to spread and thin over time, which is physically correct behavior.

2.8 Snowfall and Accumulation

When the base temperature is below 0 °C, there is a 70% probability of snowfall each month. Snowfall amount is drawn from Uniform$(0.1, 0.5)$ m of ice equivalent. Snowfall adds to ice thickness and resets the cell's surface to fresh-snow albedo (0.85), which then decays over subsequent months.

2.9 Cell States and Transition Rules

Each cell is in one of three states: `GLACIER` (ice thickness > 0), `MELTWATER` (ice just reached zero this time step), or `ROCK` (ice-free for more than one step). Meltwater cells transition to rock after one month, representing runoff.

2.10 What the Model Captures

The model captures the core BC-albedo feedback loop: soot darkens ice, which absorbs more sunlight, which accelerates melt, which concentrates soot further. It also captures seasonal forcing (temperature and solar cycles), elevation-dependent melt rates via lapse-rate correction, stochastic weather variability (temperature noise and snowfall events), spatial dynamics (edge-inward retreat, glacial flow, wind-driven soot gradients), and the distinction between fresh snow and bare ice albedo.

2.11 What the Model Does Not Capture

The model does not simulate debris cover, which insulates glaciers and slows melt as ice thins in reality. Wind direction is fixed (always east) rather than variable. There is no rain washout of soot, no algal darkening, no crevasses or basal sliding, and no sublimation. The energy balance is simplified to shortwave radiation only, ignoring longwave, sensible, and latent heat fluxes. The glacier geometry is synthetic rather than based on a real DEM. Snowfall is spatially uniform rather than concentrated on windward slopes. And the soot concentration formula has no upper cap, which leads to physically unrealistic BC values at late stages of melt (discussed further in Section 7).

3 Model Verification

Before running the Monte Carlo experiment, I verified the model with a single 10-year BAU simulation to check that the basic mechanics work as expected.

Figure 1 shows the initial glacier thickness. The Gaussian dome is centered on the grid with peak thickness of 121.9 m (including the roughness perturbation). The smooth radial gradient confirms the geometry is set up correctly.

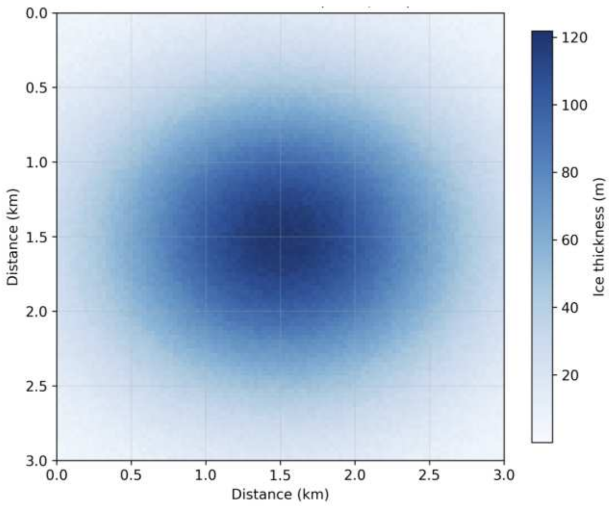

Figure 1: **Initial glacier thickness at year 0 (BAU)**. The Gaussian dome peaks at 121.9 m. All 10,000 cells are glaciated at initialization, covering the full 9.00 km^2 domain.

Figure 2 shows the glacier at years 5 and 10. By year 5, volume has declined 26.9% to 0.321 km^3 as edges begin retreating (visible as gray rock fringe). By year 10, volume is 0.183 km^3 (58.4% total loss) with evident thinning from the outside inward. The dome structure is still intact but noticeably shorter. The fact that edges melt first while the center persists is exactly what we expect from the lapse-rate correction, since edge cells are at lower (warmer) elevations.

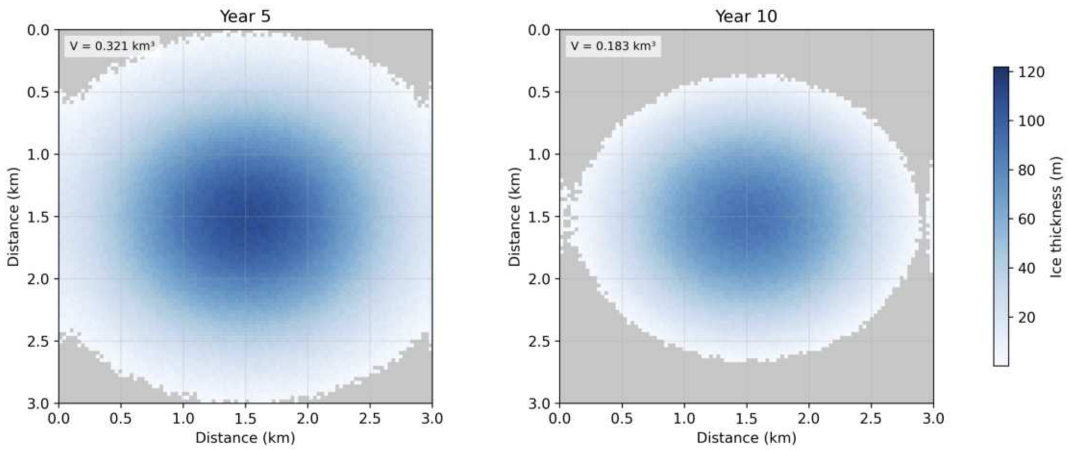

Figure 2: **Glacier thickness at years 5 (left) and 10 (right)**. Volume drops from 0.439 km^3 to 0.321 km^3 by year 5 and 0.183 km^3 by year 10. Both panels share the same color scale (max = 121.9 m). Gray areas at the edges are exposed rock, confirming edge-inward retreat.

Figure 3 shows total volume over the 10-year run. Winter accumulation increases volume, summer melt decreases it. The overall trend is declining, and the decline accelerates in later years as soot accumulates and albedo drops.

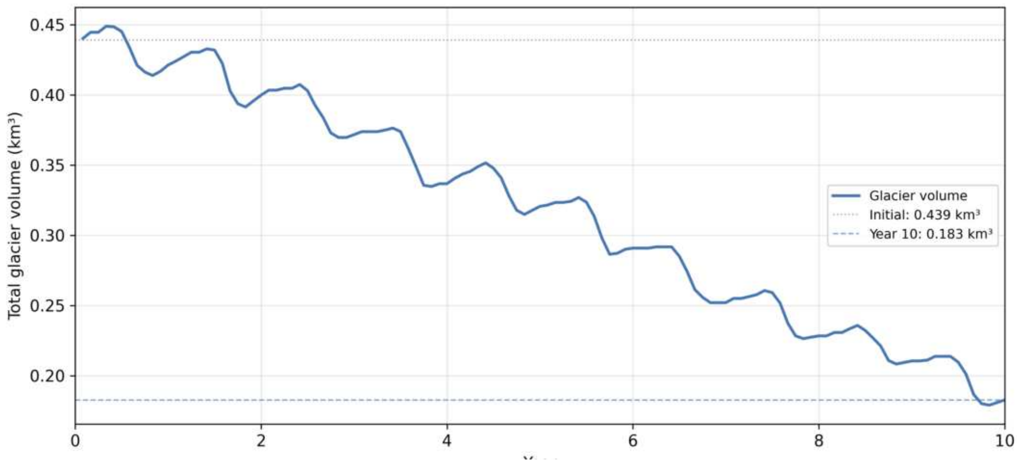

Figure 3: **Total glacier volume over 10 years (BAU)**. The sawtooth pattern reflects seasonal accumulation and melt cycles. Volume loss accelerates after year 5 as the soot concentration feedback kicks in.

Figure 4 shows mean albedo over the same period. Albedo starts near 0.71 (between fresh snow and bare ice) and declines to 0.279 by year 10 as soot builds up. Seasonal spikes reaching 0.55–0.71 correspond to snowfall events that temporarily reset the surface. Out of 120 months, 49 had snowfall, closely aligned with the 70% probability applied to roughly 6 cold months per year (expected value: $0.7 \times 6 \times 10 = 42$ events, so 49 is reasonable given randomness).

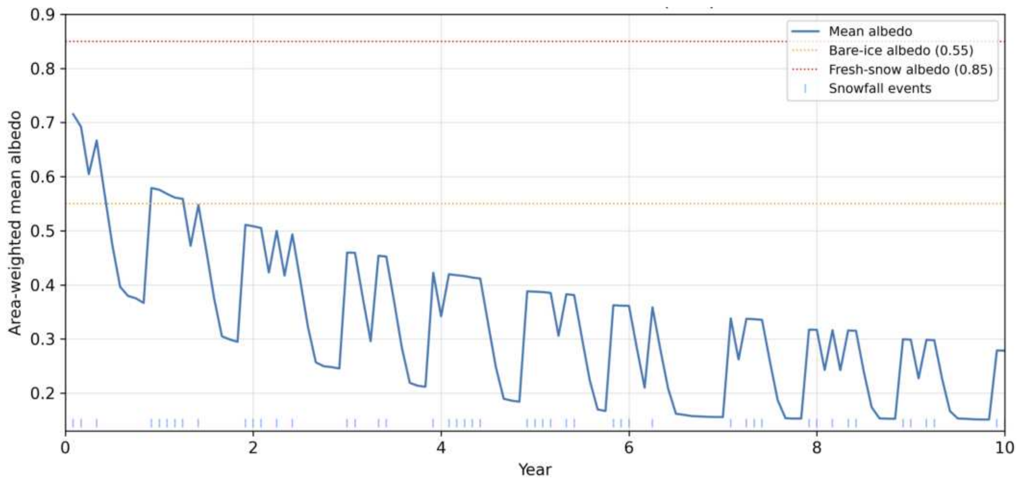

Figure 4: **Mean surface albedo over 10 years (BAU)**. Snowfall events produce seasonal spikes. The downward trend reflects BC accumulation. By year 10, mean albedo is 0.279, well below the bare-ice value of 0.55.

Figure 5 shows mean BC concentration. Concentration is 60.6 ppb at year 1, 926.9 ppb at year 5, and 23,221 ppb by year 10. The rapid growth is driven by the concentration feedback: as cells thin, the same soot mass sits on less ice, so concentration increases nonlinearly. A spike near year 9.5 reaches roughly 225,000 ppb as the last thin cells approach zero thickness. These late-stage concentrations are physically unrealistic (observed values on real glaciers are 5–500 ppb), which is a known limitation of the model discussed in Section 7.

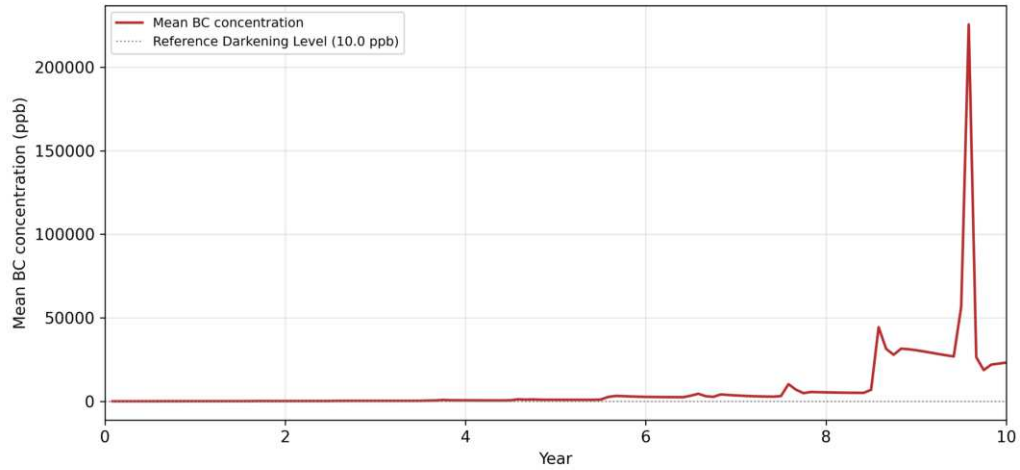

Figure 5: **Mean BC concentration over 10 years (BAU)**. Concentration grows slowly until roughly year 4, then accelerates as thinning ice concentrates deposited soot. The spike near year 9.5 (roughly 225,000 ppb) coincides with the final thinning of the last glacier cells. These late-stage values are physically unrealistic.

All five verification checks confirm the model behaves as intended: the Gaussian geometry initializes correctly, melt proceeds from edges inward due to lapse-rate cooling, albedo declines as soot accumulates, snowfall events temporarily restore albedo, and volume loss accelerates due to the positive feedback loop.

4 Monte Carlo Experiment and Results

4.1 Experiment Design

I ran 50 paired simulations per strategy over 40 years (480 monthly time steps each), for a total of 100 simulation runs. The pairing means that BAU run k and Scrubbers run k share the same random seed k, so they experience identical temperature noise and snowfall sequences. The only difference between paired runs is the BC deposition rate. This paired design isolates the emission strategy effect and reduces variance in the comparison.

4.2 Volume and Area Trajectories

Figure 6 shows mean glacier volume over 40 years with 95% confidence intervals (CIs). Both strategies start at 0.439 km^3 and decline toward zero. The trajectories diverge from approximately year 3 onward. At year 10, BAU retains 0.210 km^3 while Scrubbers retains 0.263 km^3, a 25% advantage. At year 20, the advantage grows to 82% (0.069 vs. 0.038 km^3). The CI bands are very narrow throughout, indicating low run-to-run variance.

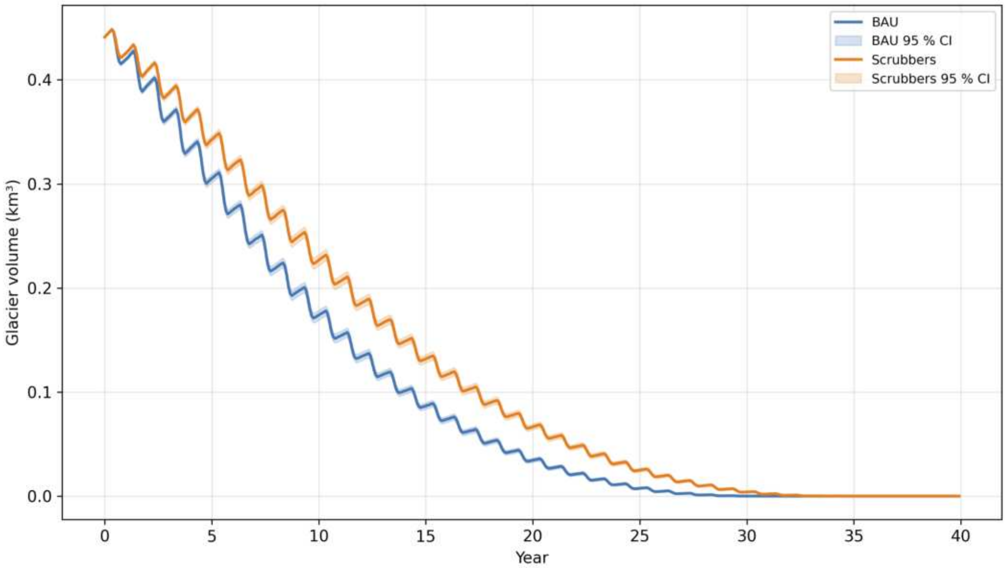

Figure 6: **Mean glacier volume with 95% CI across 50 MC runs per strategy**. BAU (blue) and Scrubbers (orange) diverge from year 3 onward. Scrubbers retains 25% more volume at year 10 and 82% more at year 20. Both reach zero within 34 years. The seasonal sawtooth is visible in both trajectories.

Figure 7 shows glacier area over time. Both start at 9.00 km^2. At year 10, BAU retains 5.64 km^2 vs. Scrubbers 6.52 km^2 (15.6% more). BAU area reaches zero around year 31, Scrubbers around year 35. The stepped decline reflects the discrete CA grid where area drops in cell-sized increments.

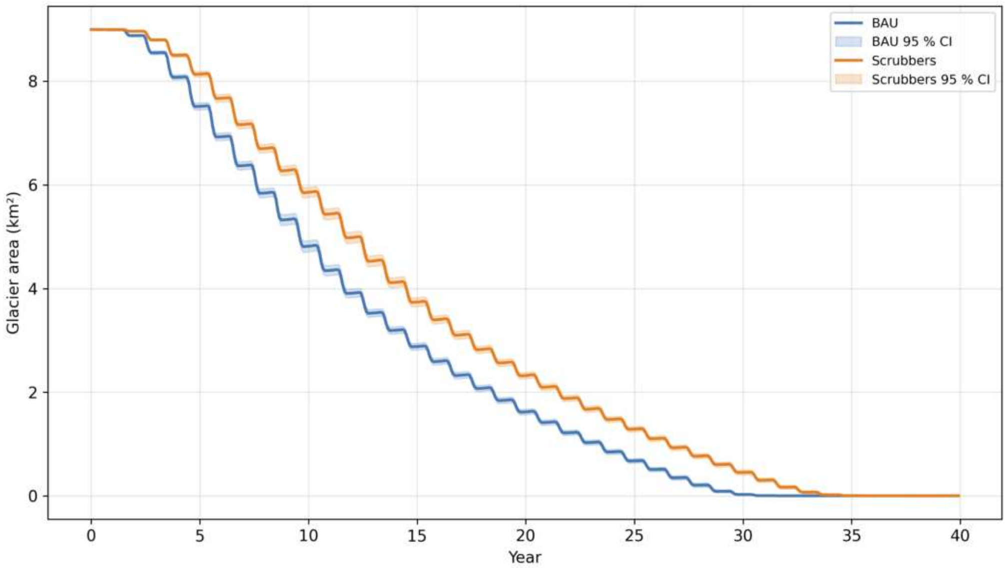

Figure 7: **Mean glacier area with 95% CI**. Scrubbers retains 15.6% more area at year 10. BAU loses all area by year 31, Scrubbers by year 35. The step pattern comes from discrete cell extinction events on the 100×100 grid.

4.3 Albedo and Meltwater Trajectories

Figure 8 shows mean albedo over time. Scrubbers maintains albedo 0.02–0.05 higher than BAU from year 1 onward. BAU converges to the 0.15 rock floor by approximately year 22, Scrubbers by year 30 as the glacier disappears entirely.

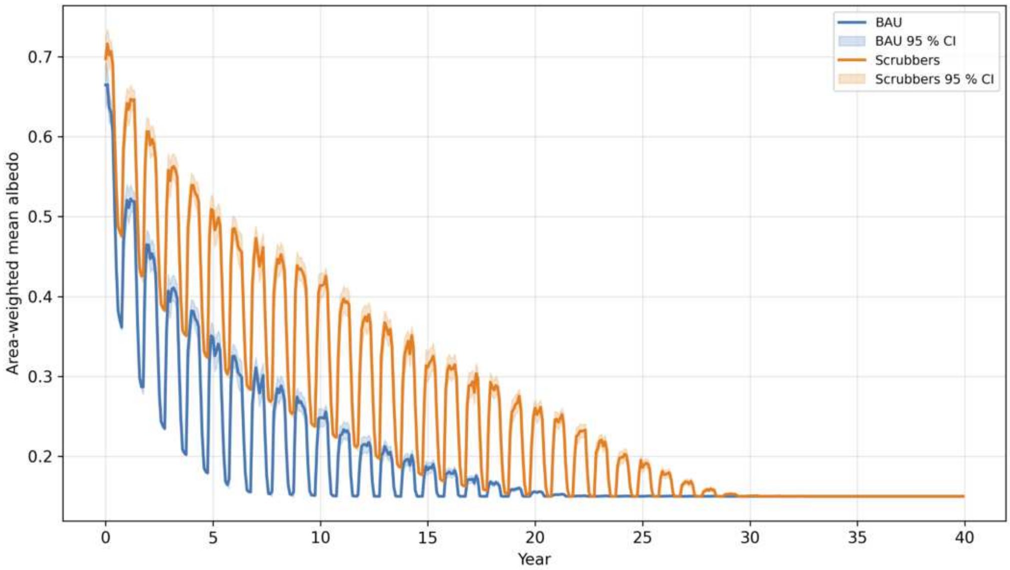

Figure 8: **Mean surface albedo with 95% CI**. Scrubbers maintains higher albedo throughout the simulation. Both converge to the 0.15 rock floor as the glacier disappears. The dense oscillations are seasonal snowfall events.

Figure 9 shows monthly meltwater discharge. BAU produces higher summer peaks (mean peak across runs: 17.3 Mm^3) than Scrubbers (14.8 Mm^3). Both converge to zero by year 35. The peak discharge is relevant for downstream flood risk.

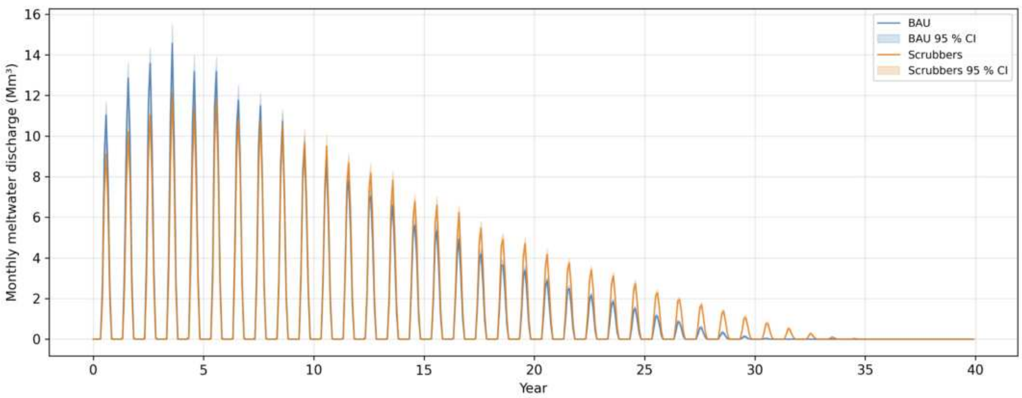

Figure 9: **Monthly meltwater discharge across 50 MC runs**. BAU peaks at 14–15 Mm^3 in years 2–5 and produces 16.9% higher mean peak discharge than Scrubbers (17.3 vs. 14.8 Mm^3). Both converge to zero as the glacier disappears.

4.4 Glacier Lifespan Distributions

Glacier lifespan is defined as the year when total volume first drops below 10% of the initial value. Figure 10 shows the distribution across 50 runs per strategy.

BAU lifespans cluster between 17 and 20 years with a mean of 18.8 years. Scrubbers spans 20–24.5 years with a mean of 22.4 years. The difference in means is 3.6 years. Neither strategy produces any survivors (runs where the glacier retains more than 10% of its volume at year 40).

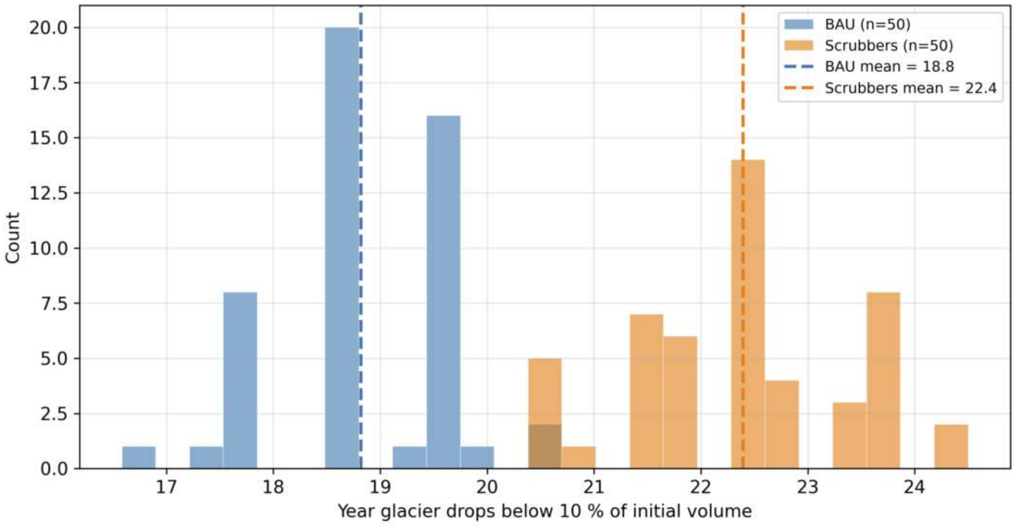

Figure 10: **Distribution of glacier lifespans (year volume drops below 10% of initial)**. BAU mean: 18.8 yr (range 16.5–20.0). Scrubbers mean: 22.4 yr (range 20.0–24.5). An 80% BC emission reduction extends glacier survival by 3.6 years on average. No survivors in either strategy after 40 years.

Figure 11 shows the distribution of peak single-month meltwater discharge. BAU produces higher peaks on average (17.3 vs. 14.8 Mm^3), but the distributions overlap substantially, meaning some Scrubbers runs produce higher peak discharge than some BAU runs due to stochastic weather variability.

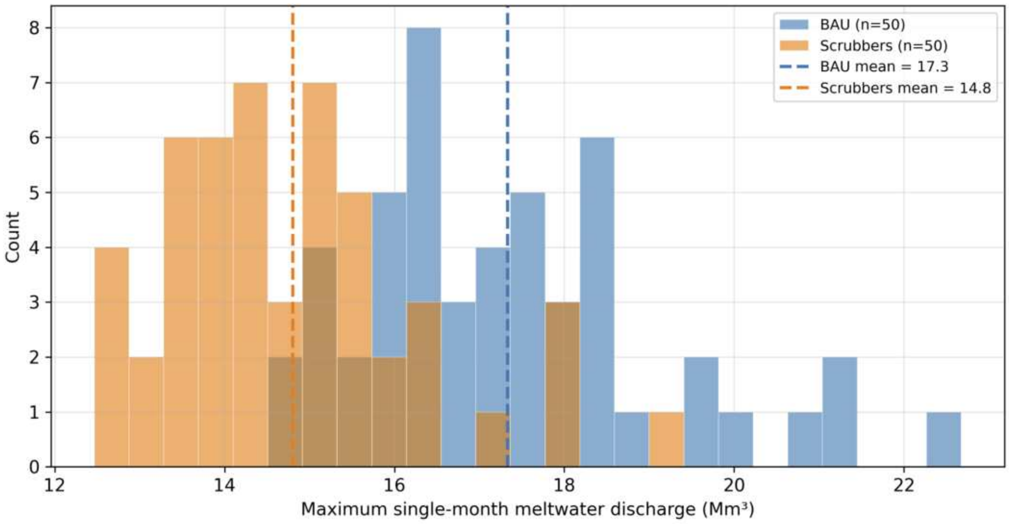

Figure 11: **Distribution of peak monthly meltwater discharge**. BAU mean: 17.3 Mm^3 (range 14–23). Scrubbers mean: 14.8 Mm^3 (range 12–19). BAU produces 16.9% higher mean peak discharge, but the distributions overlap due to weather variability.

4.5 Spatial Evolution

Figures 12–14 show side-by-side glacier thickness maps at years 10, 20, and 30 from a representative paired run.

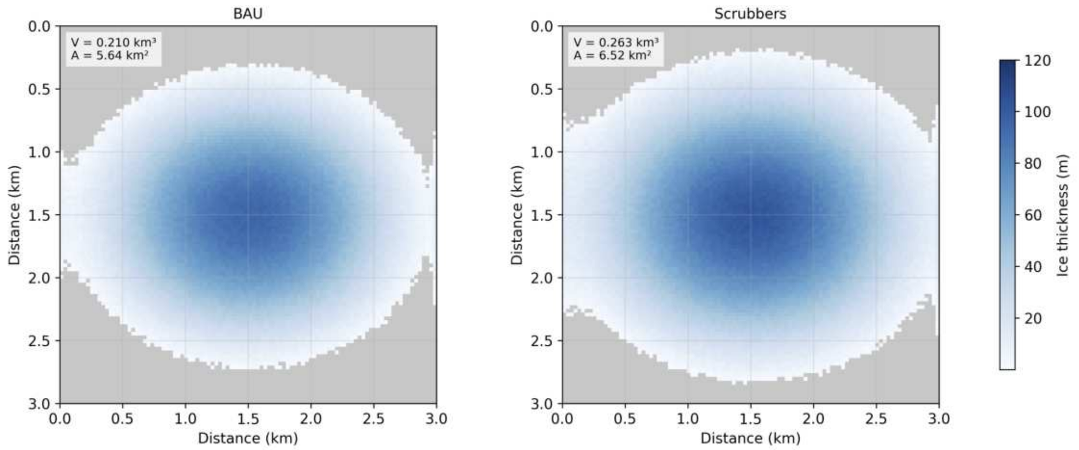

Figure 12: **Glacier thickness at year 10, BAU (left) vs. Scrubbers (right)**. BAU retains 0.210 km^3 over 5.64 km^2; Scrubbers retains 0.263 km^3 over 6.52 km^2. Both domes are intact but BAU shows more edge retreat.

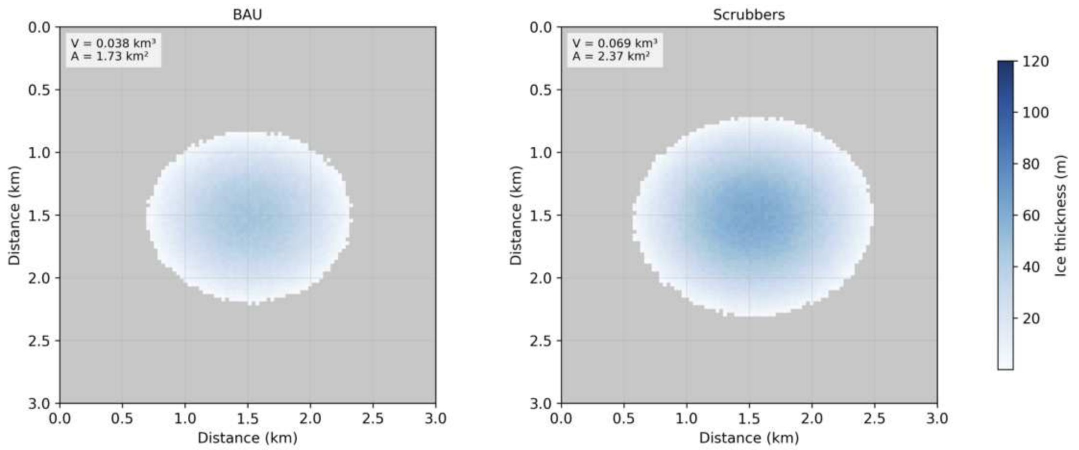

Figure 13: **Glacier thickness at year 20, BAU (left) vs. Scrubbers (right)**. BAU is near extinction (0.038 km^3, 8.7% of initial) while Scrubbers retains 0.069 km^3 (15.7%). Both glaciers are reduced to thin remnant patches at the domain center.

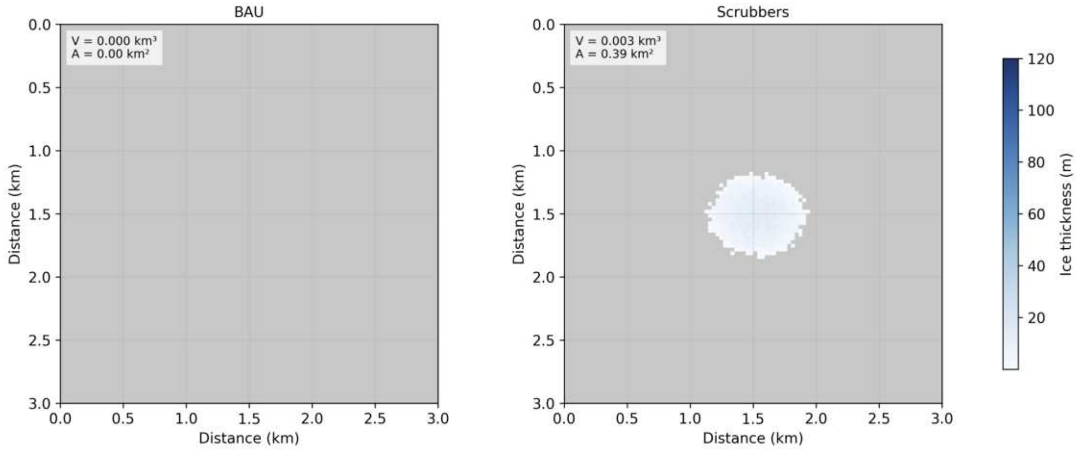

Figure 14: **Glacier thickness at year 30, BAU (left) vs. Scrubbers (right)**. BAU is completely extinct ($0.000\,\mathrm{km}^3$, fully gray rock). Scrubbers retains a tiny $0.003\,\mathrm{km}^3$ patch across $0.39\,\mathrm{km}^2$. The contrast between the two panels visually summarizes the effect of emission reduction.

The spatial evolution confirms that retreat proceeds from edges inward, consistent with lapse-rate cooling protecting the thick dome center. By year 30, the BAU glacier is gone while the Scrubbers glacier has a small remnant, buying roughly 6–7 additional years before total disappearance.

4.6 Animations

Three animations were generated showing the month-by-month glacier evolution:

- BAU melt animation (40 years): melt_bau.gif
- Scrubbers melt animation (40 years): melt_scrubbers.gif
- Side-by-side comparison: melt_comparison.gif

5 Theoretical Analysis

5.1 Mean-Field Approximation

To understand the simulation dynamics analytically, I derive a mean-field model that treats the glacier as a single spatially-uniform cell. Instead of tracking 10,000 individual cells with their own temperatures, soot levels, and albedo values, the mean-field model collapses everything into two coupled ordinary differential equations for total volume $V(t)$ and average BC concentration $C(t)$.

5.2 Deriving the ODEs

The initial volume V_0 is computed from the continuous Gaussian integral:

$$V_0 = H_{\text{peak}} \cdot \pi \cdot \sigma_x \cdot \sigma_y \cdot A_{\text{cell}} \tag{12}$$

where $A_{\text{cell}} = 30^2 = 900\,\mathrm{m}^2$. The melt-season temperature is taken as the peak summer value with the warming trend applied:

$$T_{\text{eff}}(t) = \max(0,\ T_{\text{summer}} + r_{\text{warm}} \cdot t) \tag{13}$$

where $T_{\text{summer}} = T_{\text{base}} + A_{\text{season}} \cdot \sin(2\pi(6-4)/12) \approx 6.66\,^{\circ}\text{C}$ at year 0. The albedo is computed from $C(t)$ using the same log-linear formula as the simulation (Equation 6), but applied to the spatially-uniform concentration.

The melt constant k_{melt} is derived from the energy balance equation with a 6-month melt season approximation:

$$k_{\text{melt}} = \frac{S_{\text{summer}} \cdot 86400 \cdot 30 \cdot 6}{\rho_{\text{ice}} \cdot L_f \cdot T_{\text{scale}}} \tag{14}$$

where $S_{\text{summer}} = S_{\text{mean}} + S_{\text{amp}} \cdot \sin(2\pi(5-3)/12) \approx 330\,\text{W/m}^2$ is the peak summer solar flux. Annual accumulation is estimated as the expected snowfall over 6 cold months:

$$\text{Acc} = 6 \cdot P_{\text{snow}} \cdot \frac{h_{\min} + h_{\max}}{2} \cdot A_{\text{total}} \tag{15}$$

The system of equations is then:

$$\frac{dV}{dt} = -k_{\text{melt}} \cdot (1 - \alpha(C)) \cdot T_{\text{eff}}(t) \cdot A_{\text{total}} \cdot \frac{V}{V_0} + \text{Acc} \tag{16}$$

$$\frac{dC}{dt} = D + C \cdot \frac{\max(0, -dV/dt)}{V} \tag{17}$$

where D is the mean monthly BC deposition rate (5.0 ppb/month for BAU, 1.0 for Scrubbers), and the V/V_0 factor scales the melt rate down as the glacier shrinks. The second equation captures the concentration feedback: when ice melts ($dV/dt < 0$), soot concentrates in proportion to the volume loss rate.

The system is integrated from $t = 0$ to $t = 40$ years using SciPy's RK45 solver with monthly output resolution.

5.3 Empirical vs. Theoretical Comparison

Figure 15 compares the mean-field solution (dashed line) with the Monte Carlo ensemble mean and 95% CI (solid line with shading) for both strategies.

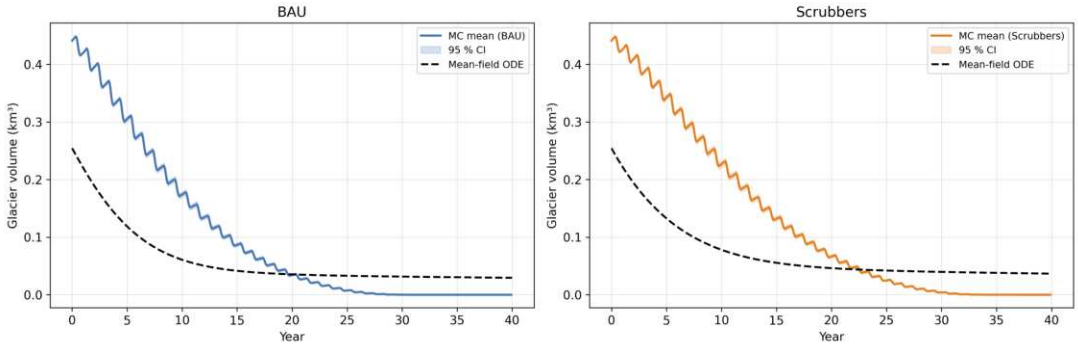

Figure 15: **MC ensemble mean (solid, with 95% CI) vs. mean-field ODE (dashed) for BAU (left) and Scrubbers (right)**. The mean-field predicts faster early melt than MC (RMSE: $0.096\,\text{km}^3$ for BAU, $0.111\,\text{km}^3$ for Scrubbers; bias: -0.044 and $-0.057\,\text{km}^3$). At late times, the mean-field predicts a small non-zero residual while MC reaches zero.

The mean-field model lies below the MC mean through approximately year 15 (negative bias of $-0.044\,\text{km}^3$ for BAU and $-0.057\,\text{km}^3$ for Scrubbers). After that, the MC runs all reach zero while the mean-field asymptotes to a small residual of 0.030–$0.037\,\text{km}^3$. The RMSE is $0.096\,\text{km}^3$ for BAU and $0.111\,\text{km}^3$ for Scrubbers.

5.4 Why the Mean Field Diverges from Monte Carlo

The mean-field model overestimates early melt because it ignores spatial variation. In the simulation, the center of the glacier is colder than the edges (because of the lapse-rate correction), and the center is also where most of the ice volume sits. So the thick, cold center melts slowly while the thin, warm edges melt fast. The mean-field model applies one average temperature to the entire glacier, which ends up melting the bulk of the ice too quickly. On top of that, the mean-field model assumes every cell has the same soot concentration, but in the simulation wind pushes soot eastward, so some cells stay cleaner and melt slower than the uniform average would predict.

At late times the divergence reverses. The simulation has discrete cells that individually hit zero thickness and disappear, so eventually the whole glacier reaches zero volume. The system of equations is continuous and can only approach zero without reaching it, which is why it predicts a small leftover volume (0.030–$0.037\,\text{km}^3$) at year 40 that does not exist in the simulation.

6 Confidence Interval Analysis

To determine how many MC runs are needed for reliable estimates, I ran BAU experiments with $N = 25, 50, 100, 200$ and computed the bootstrap 95% CI width on median glacier lifespan (2,000 resamples per estimate).

N (runs)	Median Lifespan (yr)	95% CI Width (yr)
25	20.8	1.00
50	21.5	0.92
100	21.6	0.08
200	21.6	0.04

Table 1: Bootstrap CI width on BAU median lifespan vs. number of MC runs. CI narrows 11× between $N = 50$ and $N = 100$, making $N = 100$ the practical convergence threshold.

Figure 16 shows the convergence visually. The CI width drops sharply between $N = 50$ and $N = 100$, going from 0.92 yr to 0.08 yr. At $N = 200$, the CI is only 0.04 yr wide. The collapse is faster than the classical $1/\sqrt{N}$ rate because the lifespan distribution is highly concentrated (standard deviation of only about 1 year), so the bootstrap CI collapses once enough samples cover the narrow distribution.

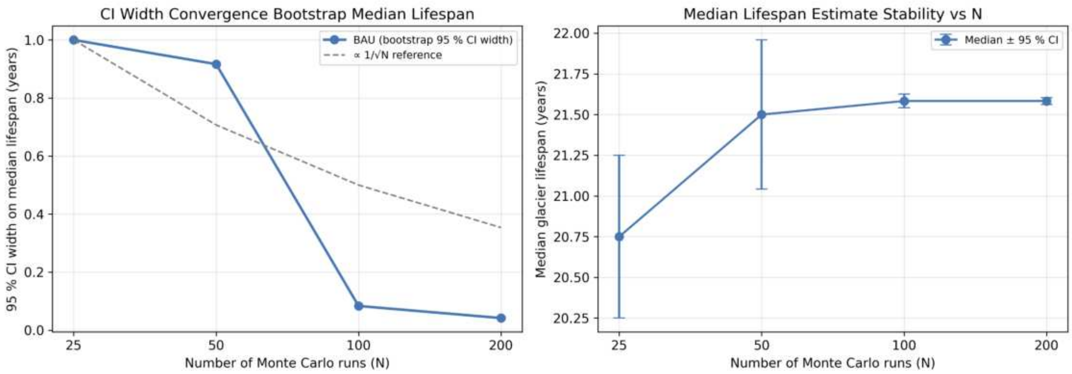

Figure 16: **Left: bootstrap 95% CI width vs. number of MC runs. Right: median lifespan estimate with error bars**. $N = 100$ is where the estimate stabilizes. The production run used $N = 50$, giving ± 0.46 yr precision on the median.

The production experiment used $N = 50$ runs per strategy, which gives a CI width of 0.92 years on median lifespan. For the strategy comparison (BAU vs. Scrubbers), the paired design reduces uncertainty because weather noise cancels out. The 3.6-year difference in mean lifespan is about 4× the CI width, so the difference is well-resolved even at $N = 50$.

The median lifespan from the CI analysis (21.6 yr) is approximately 3 years higher than from the main experiment (18.8 yr) because the CI analysis ran separate simulations with different seed sequences. Since lifespan depends on the specific weather realizations, different seed ranges can shift the median by a few years. Increasing to $N = 200$ or higher would resolve this.

7 Discussion

The main finding is that an 80% reduction in BC emissions extends glacier lifespan by 3.6 years on average (from 18.8 to 22.4 years) and reduces peak monthly meltwater discharge by 16.9% (from 17.3 to 14.8 Mm^3). Both results are consistent with what the physical literature predicts: BC deposition contributes to roughly 13–20% of glacier melt in affected regions [1, 4], so reducing BC by 80% should slow melt noticeably but not prevent it under continued warming.

Neither strategy prevents glacier loss within 40 years. Under a warming rate of 0.03 °C/yr, the temperature trend alone is enough to drive the glacier to extinction even with zero soot. The scrubbers buy time but do not change the outcome. From a policy perspective, BC reduction is a useful short-term measure (it delays peak meltwater by several years, giving downstream communities more time to adapt) but cannot substitute for addressing the underlying warming trend.

The most significant model limitation is the uncapped soot concentration feedback. By year 10 of a BAU run, mean BC concentration reaches 23,221 ppb, which is 2–3 orders of magnitude above observed values on real glaciers (typically 5–500 ppb). The formula $C_{\text{new}} = C_{\text{old}} \cdot h/(h - \Delta h)$ causes exponential increase as cells approach zero thickness. In reality, fresh snowfall buries and dilutes surface soot, rain washes it away, and wind dispersal removes it. A future improvement would be to add a BC decay or dilution term, or to cap concentration at a physically realistic maximum (e.g., 500 ppb).

The mean-field model gets the general shape of the volume decline right, but melts ice too fast early on (RMSE of 0.096–0.111 km^3) because it does not account for the cold center of the glacier where most of the volume is. At late times it predicts a small leftover volume that the simulation correctly brings to zero. It is a useful approximation for understanding the system, but not a replacement for the full spatial simulation.

8 Conclusion

I simulated the melt of a synthetic mountain glacier under two BC emission scenarios using a Monte Carlo cellular automata model. The simulation incorporates the BC-albedo feedback loop, seasonal temperature and solar forcing, stochastic weather, glacial flow, and wind-driven soot redistribution.

Reducing BC emissions by 80% via factory scrubbers delays glacier extinction by 3.6 years (from 18.8 to 22.4 years median lifespan) and reduces peak monthly meltwater discharge by 16.9% (17.3 to 14.8 Mm^3). Under a warming rate of 0.03 °C/yr, neither strategy prevents glacier loss within 40 years. The mean-field model captures the broad volume decline but underestimates volume by 0.04–0.06 km^3 on average due to the spatial heterogeneity the CA captures and the mean-field approximation ignores.

For a regional water authority managing a glaciated watershed, installing scrubbers is worthwhile because it buys 3–4 years of additional glacier-fed water supply and reduces peak flood discharge by 17

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